

A Coherence-Gated Symmetron Model with Axial Spin Coupling Near Current Fifth-Force Bounds

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Abstract

We propose a modification of symmetron scalar field models in which the effective coupling to matter is enhanced by macroscopic axial spin coherence. The model introduces a relativistic axial-vector coupling between the scalar field and fermionic spin currents, leading to a continuous, state-dependent effective interaction strength. We derive the resulting Yukawa-type force law and identify a narrow parameter space near existing experimental bounds where the model remains viable.

1 Introduction

Scalar field extensions to gravity such as symmetron models introduce screening mechanisms that allow additional forces to evade detection in high-density environments. We explore a modification where coupling depends on macroscopic spin coherence.

2 Theory

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{1}{2}(\partial_\mu \phi)^2 - V(\phi) + \frac{\phi^2}{2M^2} T^\mu_\mu + g_s \phi \bar{\psi} \gamma^\mu \gamma^5 \psi n_\mu \right] \quad (1)$$

$$V(\phi) = -\frac{1}{2}\mu^2 \phi^2 + \frac{1}{4}\lambda \phi^4 \quad (2)$$

3 Force Law

$$F(r) = -\frac{Gm_1 m_2}{r^2} \left[1 - \alpha_{\text{eff}} \left(1 + \frac{r}{\lambda} \right) e^{-r/\lambda} \right] \quad (3)$$

4 Coupling

$$\alpha_{\text{eff}} = \frac{\phi_{\text{vac}}^2}{M^2} \left(\frac{\mathcal{O}}{\mathcal{O}_{\text{max}}} \right) \quad (4)$$

5 Conclusion

This model introduces a coherence-dependent extension to symmetron scalar field theories and motivates experimental investigation.